METALS AND METALLIC BONDING 13

Metals can be recognised by their characteristic physical properties.

1 Name three typical properties of metals.

Metals form two types of bonds: ionic bonds and metallic bonds. They form ionic bonds with non-metals. In ionic bonding, metals obtain noble gas electron structure by transferring electrons to non-metals.

2 What type of ions are formed by metals in ionic bonds?

In metals, the bonding between atoms is often strong. Metal structures contain billions of atoms packed together very tightly. The atoms are packed together in simple structures. These patterns can be modelled by dropping many marbles into a small box. Many metals are dense; they have atoms that are heavy and pack closely together. When metal atoms bond together, they do not form a noble gas type structure.

3 Explain what is meant by *noble gas type structure*.

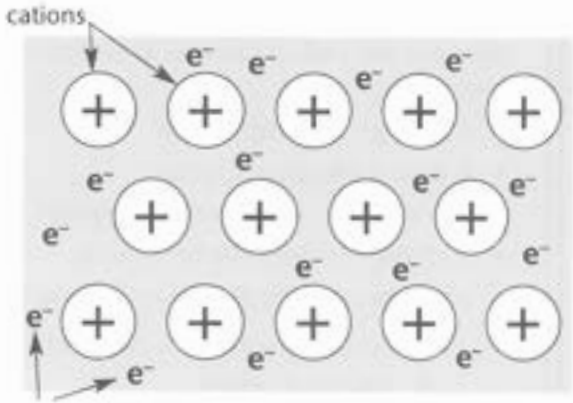


Figure 1

Electrons are still shared in metallic bonds, but these bonds are different from covalent bonds in one important way. In a covalent bond, the electrons spend their time between the two atoms that are bonded. When metal atoms share their electrons, these electrons are free to move anywhere within the metal structure. There is a 'sea of electrons' surrounding the positive metal cations (see figure 1). Another way to visualise this is to imagine metals as three-dimensional lattices of positive ions in a 'sea' of valence electrons. The delocalised outer-shell electrons in the lattice are extremely mobile and can readily move through the lattice when excited by either thermal or electrical energy. Metallic bonding is strong; hence metals are often hard and have high melting points. However, layers in the metal lattice can be forced to 'slip over' each other; this accounts for the malleability and ductility of metals. Mixtures of different metals are called alloys.

4 What are *valence* electrons?

5 Give another name for *thermal energy*.

6 Define *ductile*.

7 Explain why metals are good electrical conductors.

8 Use figure 2 to describe how the electrical conductivity of a metal varies with temperature.

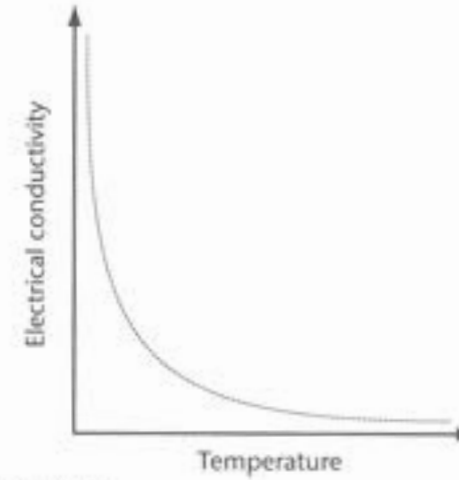


Figure 2

9 Match the following terms with the definitions below:
alloy, cation, lattice, malleable, metallic bonding

a The electrostatic force of attraction that neighbouring cations have for the delocalised electrons surrounding them: _____

b A substance made up of two or more metals or a metal and a non-metal mixed together, usually when molten: _____

c An atom or group of atoms that carries a positive electric charge: _____

d Can have its shape changed by beating with a hammer or by the pressure of rollers: _____

e The regular geometrical arrangement of particles in a metal or crystal: _____

10 Metals can form two types of bonds: metallic and ionic. Explain the differences between these types of bonds.

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